

AI-Assisted Recruitment Analytics

Approach & Methodology

Project Overview

This project was developed to analyze a fictional recruitment dataset and transform it into actionable business insights. The objective was not only to identify recruitment metrics, but also to demonstrate a structured analytical approach supported by AI, transforming raw data into business recommendations. To explore different ways of communicating the same analysis, two complementary visualization formats were created: an Executive Excel Dashboard focused on business KPIs and decision-making, and an Interactive HTML Knowledge Base designed as a second, AI-generated visualization that reorganizes the same analytical content into a more interactive and reusable experience. Both visualizations are based on the same dataset and analytical conclusions, but each serves a different purpose.

Approach

The analysis began with understanding the dataset structure, recruitment stages and available variables. The data was then prepared by formatting it as an Excel table, checking for duplicate candidates, reviewing missing values and validating potential inconsistencies. The exploratory analysis covered the overall recruitment funnel, funnel by role, source performance, recruiter performance, candidate signals, response time, rejection reasons and qualitative recruiter notes. Rather than producing isolated charts, the findings were consolidated into a business narrative supported by recommendations.

Decisions Made

The recruitment funnel was rebuilt for each role to verify whether the main bottleneck was process-wide or role-specific. Recruiter performance was interpreted considering portfolio complexity rather than conversion alone. A dedicated response-time analysis was added after identifying that candidates who responded several days later still achieved competitive hiring rates. After completing the Excel dashboard, a second visualization was intentionally created using AI. Instead of reproducing the dashboard, the HTML version reorganizes the same analytical content into an interactive knowledge base, demonstrating how AI can transform one analytical foundation into different user experiences without changing the underlying business logic.

Use of AI

AI was used as an analytical copilot throughout the project. It supported dataset understanding, analytical planning, hypothesis generation, role-level funnel segmentation, organization of qualitative recruiter notes, consistency review and storytelling refinement. AI was also used to generate the second visualization of the project—the Interactive HTML Knowledge Base—showing how the same insights can be repurposed into a different, more interactive experience while preserving analytical consistency. All findings and recommendations were validated against the original dataset before inclusion.

Key Insights

- The recruitment process performs consistently throughout the early stages of the funnel.
- The largest opportunity lies in the Technical Test → Offer transition across every analyzed role.
- 'No Response' is the primary rejection reason, highlighting opportunities to improve sourcing outreach and candidate nurturing.
- Response time alone is not a reliable stop signal, as candidates replying after several days still achieved competitive hiring rates.
- GitHub and Inbound delivered the highest hiring conversion rates.
- Hired candidates tend to combine multiple positive indicators instead of a single defining characteristic.
- Improving ATS data quality will increase analytical reliability and enable more advanced AI-supported recruiting workflows.

Final Consideration

Creating a second visualization was a deliberate decision to demonstrate that AI can extend beyond data analysis, enabling the same analytical foundation to be communicated through different experiences tailored to different audiences and use cases.